INCH-POUND

MIL-G-80073C
30 September 1988
SUPERSEDING
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25 July 1973

MILITARY SPECIFICATION

GRINDING MACHINES, INTERNAL, CHUCKING, UNIVERSAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers universal, internal, chucking, horizontal spindle grinding machines with manual and automatic sizing with equipment and accessories specified herein.

1.2 Classification. The machine shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 3.5 and 6,2.1).

Type I - Manual sizing with manual/power operation.
Type II - Automatic sizing with manual/power operation.

Size	Swing (inside standard chuck guard), inches *	Maximum grinding stroke, inches
1	16	16
2	20	16

* May be increased using spacer or riser blocks.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational 'Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of DOCUments, Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.8 - Chucks and Chuck Jaws. (DOD adopted)

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools, (DOD adopted)

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22102-4269.)

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NATIONAL ELECTRICAL Manufacturers ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers? Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3,2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of precision internal, external, taper, and rotary surface grinding in accordance with the requirements herein. The operator shall control grinding the workpiece to size on type I machines, Type II machines shall automatically grind workplaces to predetermined sizes. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair.

3.2.1 Design and construction measurement systems. Unless otherwise specified (see 6.2.1), either inch-pound units (see 6.4.1) or metric units (see 6.4.2) shall be used in the design and construction of the machine.

3.2.1.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound, metric, or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

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3.2.1.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch for coarse feed and 0.0001 inch for fine feed dials. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize its intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Mercury restriction, The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be accessible.

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3.2.9 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as weldings, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturers commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable ‘Detailed Standard” section referenced therein.

3.3.7 Gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine, Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

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3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461, The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, workhead slide assembly, workhead, wheelhead slide assembly, wheelhead, driving mechanisms, controls, and wheel truing device. Any other components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a floor-mounted, box-type, stress-relieved casting of iron or iron alloy or weldment, ribbed and braced to withstand forces imposed when the machine is used to its maximum capacities. The base shall be provided with hardened and ground or scraped precision ways or a hydrostatic way system to support and guide wheelhead assembly movement and a precision machined surface for mounting the workhead slide assembly. Machine ways shall be protected wherever possible to exclude grinding dust, dirt, coolant, and other hazards. Hydraulic and coolant reservoirs shall be housed in the base or separate external units. Access doors or removable plates shall be provided, as necessary, to service components housed within the base. Means shall be provided to level and secure the machine to the floor.

3.4.2 Workhead slide assembly. The workhead slide assembly shall have way surfaces mating to the corresponding base ways. The slide shall allow adjustment of the workhead in two directions for various workpiece diameters and lengths. The slide top shall support the workhead and its components and shall permit workhead swivel to facilitate taper and rotary face grinding. The slide shall be positioned manually using a handwheel with a dial graduated in increments of not more than 0.001 inch. When specified (see 6.2.1), the slide shall be positioned by power. Means shall be provided to lock the slide after positioning.

3.4.3 Workhead. The workhead shall mount on the workhead slide assembly. The workhead shall be oiltight and shall house the workhead spindle and related components. The workhead shall swivel from 0 degrees (parallel to the Z axis straight bore grinding position) to 90 degrees to the rear to permit taper bore, face, and rotary surface grinding. A graduated scale, with increments not larger than 1 degree, and index marker shall be provided to indicate workhead angular setting for the 0 degrees to 90 degrees angular range. Means shall be provided to lock the workhead after positioning.

3.4.3.1 Workhead spindle. The workhead spindle shall be steel alloy, hardened to not less than Rockwell C-45, ground, and mounted in precision, preloaded, antifriction ball or roller bearings. The workhead spindle bearings shall be protected from contamination. The spindle shall be continuously or permanently lubricated and provide vibration-free operation at all speeds. The spindle shall be a flange-style spindle accepting a chuck or other workholding fixture at one end and a driving mechanism at the other end. Unless otherwise specified (see 6.2.1), the spindle flange shall be manufacturer's standard.

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3.4.3.2 Workhead spindle drive. The workhead spindle drive shall include a reversible, infinitely variable hydraulic or DC drive motor and belt and pulley arrangement. When specified (see 6.2.1), a speed indicating device calibrated in revolutions per minute (RPM) over the applicable speed range shall be provided. Unless otherwise specified (see 6.2.1), means for quick, shockless braking of the workpiece shall be provided. Means shall be provided to maintain belt tension unless a nontensioning belt is used.

3.4.4 Wheel slide assembly. The wheel slide assembly shall have ways mated to the corresponding base ways. The wheel slide assembly shall be composed of two independent slides, a cross feed slide to provide wheelhead infeed and a longitudinally traversing table to provide longitudinal reciprocation of the wheelhead, which shall travel perpendicular to each other. Manual operation, using handwheels, of all slides shall be provided to allow for workpiece set-up and facing operations. When specified (see 6.2.1), power operation of the cross feed slide shall be provided to allow automatic infeed of the wheelhead during the grinding stroke. Automatic feed shall be at the left hand table reversal only and at both ends of table travel. When specified (see 6.2.1), automatic feed shall be provided at the left hand table reversal only, at the right hand table reversal only, and at both ends of table travel. Power operation of the table shall be provided for automatic table reversal during grinding. The length of table stroke shall be infinitely variable within the total grinding stroke of the machine. Adjustable stops shall be provided for set-up to prevent the wheel from hitting workpiece shoulders or the bottom of the workpiece bore. Wheel dwell shall be independent and adjustable at both ends of table stroke. Means shall be provided to disengage the handwheel for manual operation of the table slide when the slide is under power operation.

3.4.5 Wheelhead. The wheelhead shall be mounted on the upper wheelhead assembly slide. The number, type, style, and size of each wheelhead required shall be as specified (see 6.2.1). The sizes and styles of quills or sleeves to fit the wheelhead(s) shall be as specified (see 6.2.1).

3.4.5.1 Wheelhead spindle, The grinding wheel spindle shall be hardened and ground steel alloy, horizontally mounted in precision, preloaded, antifriction ball or roller bearings. The spindle shall be continuously or permanently lubricated and provide vibration-free operation at all spindle speeds. The spindle shall be dynamically balanced. The spindle shall accept a balanced, pre-mounted grinding wheel at one end and a driving mechanism at the other end.

3.4.5.2 Wheelhead spindle drive. Wheelhead spindle rotation shall be accomplished through a direct drive motor or a motor and belt and pulley arrangement. Motor speed shall be constant, When a belt and pulley arrangement is supplied, means shall be provided to adjust and maintain belt tension. All belts required for specified wheel speeds shall be furnished.

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3.4.5.3 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be properly guarded to protect the wheels during workpiece loading or inplace workpiece inspection. When specified (see 6.2.1), an adjustable, automatically raised and lowered wheelguard shall be provided. Wheels shall pre-mounted and balanced.

3.4.6 Controls. All operating controls shall be located convenient to the operator's work station(s). Unless otherwise specified (see 6.2.1), the following controls shall be provided on each machine:

a. Coarse feed to set feed amount and rate from 0.0001 inch to 0,0006 inch per impulse.
b. Fine feed to set feed amount and rate to be ground at 0.0001 inch per impulse.
co Means to set feed impulse at the left end, the right end, or both ends of grinding stroke, as applicable (see 3.4.4).
d. Wheel wear compensation.
e. Manual feed to feed wheel manually in 0.0001 inch increments.
f. Wheel table slide speed for grinding and dressing.
g. Adjustable wheel dwell,
h. Adjustable sparkout time.
i. Workhead spindle speed and direction of rotation.

3.4.6.1 Automatic cycle. Each type II machine shall be provided with a system to permit automatic grinding of workplaces to predetermined dimensions. The system shall allow for roughing and finishing of bores and dressing of wheels automatically. The operator shall only be required to load the workpiece, initiate the automatic cycle, and unload the workpiece, Size control shall be by diamond sizing or in-process gaging or a combination of the two processes. The cycle shall include, but not be limited to, the following:

a, New wheel dress.
b. Rapid traverse positioning of the grinding wheel.
c. Power traverse rate (reciprocation cycle) of table.
d. Rapid advance of the grinding wheel to start feed position.
e. Start/stop, direction of rotation, and speed of workhead.
f. Start/stop and flow of coolant.
g. Coarse and fine axis feed.
h. Wheel dress and compensation.
i. Sparkout,
j. Rapid axis retraction.

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3.4.6,2 Digital readout (DRO) system. When specified (see 6.2.1), a DRO system shall be provided to visually display the cross feed position and table position and actual amounts and rates of grinding feed for set-up convenience and in-cycle inspection. Rough and finish grinding feed rates and amounts shall be set separately. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be 0.0001 inch/foot, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0002 inch, measured over the full travel on each axis. Special requirements for the DRO system components shall be as specified (see 6.2.1). The DRO shall indicate axis location in both inch-pound and metric units. The digital feed and readout shall be capable of overriding preset feed during the automatic cycle (see 3.4.6.1) in either the "in" or "out" direction of feed.

3.4.6.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and inadvertent damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles. Position feedback devices shall be easily and quickly interchangeable between like machine in any multiple machine order.

3.4.6.2.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual digits not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator’s work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine.

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3.4.7 Wheel truing device. A single point diamond type, straight wheel truing device, complete with industrial diamond, shall be provided with each machine. The truing device shall be capable of truing the outside diameter and left and right face of the grinding wheel. The dresser shall be raised and lowered manually. When specified (see 6.2.1), the truing device shall automatically raise to the "rest" position when the wheel travels forward beyond the wheel dressing position. The wheel dresser on type II machines shall be compatible with the automatic sizing system (see 3.4.6.1), with the option for dressing during cycle, at the end of cycle, or multiple times during the cycle. All hardware necessary to mount the truing device on the machine shall be provided. The dresser mount shall be adjustable for various wheel sizes.

3.4.8 Coolant system. When specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, all necessary piping, and all necessary splash guarding. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct the flow of coolant and to control the amount of coolant.

3.4.9 Hydraulic system. The hydraulic system, when applicable, shall conform to or exceed the requirements of JIC H-1. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable. Hydraulic controls shall be located in a separate compartment from electrical controls.

3.4.10 Electrical system. The electrical system shall be wired to conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase, 60 hertz (Hz) circuit. Unless otherwise specified (see 6.2.1), the machine shall be initially wired for operation on 460 volts.

3.4.10.1 Motors. Motors shall meet the requirements of 3.4.10. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-I.

3.4.10.2 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area, If a built-in lighting system is provided, the adjustable light described above shall not be required.

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3.4.10.3 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.10.4 Control circuit voltage. Auxiliary control circuits shall be isolated from the input power supply by a low-voltage transformer having a secondary voltage of 120 volts or less.

3.4.10.5 Electrical wheel load indicator. When specified (see 6.2.1), the machine shall have a meter to continuously monitor grinding forces exerted on the workpiece. The meter dial shall indicate percent of rated full-load current. The accuracy of the meter shall be within 2 percent of the full-rated load. The meter shall be clearly readable from the operator's normal position.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall meet the requirements of table I for the machine specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested using the procedures given in 4.7.2.

3.7 Performance. The machine shall grind internal and external cylindrical workpieces, internal and external tapered workpieces, and rotary workpiece faces in accordance with the requirements given in 4.7.3 and 4.7.4.

3.7.1 Additional performance features. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine.

a. One set of special wrenches, operating, and repair tools normally furnished by the manufacturer.

b. One 12-inch flange-mounted faceplate having not less than two equally spaced slots.

c. One manually operated collet chuck, capacity 1/8 inch to 1-3/8 inches, for direct mounting on type spindle nose furnished in accordance with 3.4.3.1.

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d. One four-jaw, independent chuck equipped with standard 3-step reversible top jaws fitted to the size and type workhead spindle nose flange furnished (see 3.4.3.1); the chuck and top jaws shall conform to the requirements of ANSI B5.8 for class 1, medium duty, wrench-operated chucks.

e. One chuck guard, hinged to allow access to the chuck and other workholding fixtures.

f. One set of round collets complete with container, to cover a range from 1/8 inch to 1-3/8 inches, in 1/16 inch increments.

g. One extra set of workhead drive belts.

h. One extra set of wheelhead drive belts (if applicable).

i. One set of grinding wheels normally furnished by the manufacturer.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

a. Additional grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel, as specified (see 6.2.1).

b. One steady rest, capacity of 3/4 inch to 6 inches outside diameter, complete with mounting base,

c. Sine bar attachment (for taper grinding).

d. Other optional equipment, as specified.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on instruments, control panels, and plates. All words on instruments, control panels, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate be permanently and securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the information listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or Military specification for each lubricant (if applicable)

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3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such. inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2,1), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4,4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 percent to 95 percent relative humidity.

4.7.2 Alignment tolerance test. Each machine and its components shall be tested to determine compliance with the alignment tolerance requirements of table II. Alignment tests shall be performed before and after performing the tests given in 4.7.3 through 4.7.4.4. Any machine adjustments required to comply with 3.6 during or after the performance tests shall be cause for rejection. The instruments and methods used in performing the tests shall be at the option of the manufacturer provided they accurately determine the tolerances of the machine,

4.7.3 Operational test. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.4 Performance tests. The machine shall be tested to demonstrate its ability to perform the grinding operations to the tolerances given below. Grinding wheels and workpiece material shall be at the manufacturer's option. All instruments, materials, and tools required to perform and evaluate the tests shall be furnished by the supplier. The measuring instruments used shall have evidence of calibration traceable to the National Bureau of Standards (NBS). The workpiece shall be a cylinder approximately 6 inches in diameter and 8 inches in length. The workpiece may be pre-drilled or bored to 3-9/16 inches or more. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1).

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4.7.4.1 Outside diameter grinding test. The machine shall cylindrical grind the outside diameter (not less than 3/4 of the total length of the workpiece) to a size predetermined at the time of the test. The outside dimensions of the cylinder shall be round and straight to within 0.00015 inch total indicator reading (TIR). Surface finish shall be not rougher than 8 microinch.

4.7.4.2 Internal diameter grinding test. The machine shall cylindrical grind the internal diameter (the total length of the workpiece) to a size predetermined at the time of the test. The internal dimensions of the cylindrical ground hole shall be round and straight to within 0.00015 inch TIR. Surface finish shall be not rougher than 8 microinch.

4.7.4.3 Internal taper grinding test. The machine shall taper grind an internal diameter with the workhead set at 5 degrees for 4 inches of length. The taper shall be uniform to within 0.00015 inch TIR in 4 inches. Surface finish shall be not greater than 8 microinch.

4.7.4.4 Face grinding test. The machine shall rotary face grind the workpiece with the workhead set at 90 degrees (from the 0 degree setting). The total grind shall be not less than 0.010 inch. Surface finish shall be not greater than 8 microinch.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any shop for medium production multi-purpose precision grinding. Grinding operations are to be performed on straight and tapered bores, external diameters, internal and external faces, and rotary faces. The full 90 degree horizontal plane swivel capability of the workhead allows this machine to be designated as a universal machine.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. If machine is required to be configured in a specific measurement system (inch-pound or metric), state required system (see 3.2.1).

f. System of units for measuring and indicating devices (see 3.2.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Power workhead slide travel, if required (see 3.4.2).

k. Workhead spindle flange, if different (see 3.4.3.1).

1. Workhead spindle drive speed indicating device, if required (see 3.4.3.2).

m. Quick, shockless workpiece braking, if not required (see 3.4.3.2).

n. Power operated wheelhead cross slide, if required (see 3.4.4).

0. Automatic feed of wheelhead cross slide at left hand only, right hand only, and both ends of table reversal, if required (see 3.4.4).

p. Number, type, style, and size wheelhead(s) required (see 3.4.5).

q. Sizes and styles of wheelhead quills or sleeves (see 3.4.5).

r. Automatic wheelguard, if required (see 3.4.5.3).

s. Controls, if different (see 3.4.6).

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t. Digital readout system, if required (see 3.4.6.2).

u. Specify any special digital readout system features, if required (see 3.4.6.2).

v. Automatic wheel truing device, if required (see 3.4.7).

w. Coolant system, if required (see 3.4.8).

x. Electrical system, if different (see 3.4.10).

y. Initial voltage, if different (see 3.4.10).

z. Motor, if different (see 3.4.10.1).

aa. Work light, if not required (see 3.4.10.2).

bb. Hour meter, if required (see 3.4.10.3).

cc. Electrical wheel load indicator, if required (see 3.4.10.5).

dd. Size and capacity, if different (see 3.5).

ee. Additional performance features, if required (see 3.7.1).

ff. Standard equipment, if different (see 3.8).

gg. Specify and fully describe optional equipment, if required (see 3.9).

hh. Additional grinding wheels: shank size, number of each size, type, dimensional characteristics, and composition, if required (see 3.9.(a)).

ii, Specify applicable repair parts, if required (see 3.10).

jj. Lubrication plate, if not required (see 3.11.1).

kk. Information on lubrication plate, if different (see 3.11.1).

ll. Information on nameplate, if different (see 3.11.2).

mm. Inspection conditions, if different (see 4.3).

nn. First article inspection, if different (see 4.4).

00. Quality conformance inspection, if different (see 4.5).

pp. Performance test, if different (see 4.7.4).

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6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
•	•	•	•
•	•	•	•

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD From 1423.

6.2.2.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards, Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on " " International D'Unites (S1)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-G-80073B	MIL-G-80073C
Size 1	Size 1
New	Size 2

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.10 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:

M	XXXXX-	XX	X
Prefix to indicate military specification	Specification number	Type (see 1.2)	Size (see 1.2)

6.13 Subject term (keyword) listing.

Chucking
Cylindrical
Grinding machine
Internal
Universal

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TABLE I. Sizes and capacities.

Characteristic	Type I and Type II	
	Size 1	Size 2
CAPACITY:		
Maximum swing without guard, inches	23	23-1/2
Swing inside standard guard, inches	16	20
Swing with workhead swiveled, inches	12	13
Grinding stroke range, inches	1/4 to 16	
Workhead:		
Workhead cross slide travel, inches	15-1/2	17-1/2
Maximum workhead angular swivel, degrees	90	
Workhead spindle flange diameter, inches	6	
Wheelhead:		
Maximum longitudinal table travel, inches	20	
Wheelhead crossfeed travel, inches	1-1/2	1-3/8
SPEEDS AND FEEDS:		
Table grinding speed range, FPM	0 to 25	0 to 18
Table rapid traverse, FPM	25	18
Wheelhead spindle speed, maximum, RPM	3600	
Workhead spindle speed range, RPM	50 to 425	0 to 500
MOTORS:		
Workhead drive	1-1/2	
Wheelhead drive	5	
**Hydraulic pump	1-1/2	
**Coolant pump	1/2	

* Unless otherwise specified by the procuring activity (see 3.5).

**When applicable.

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size shown in the above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated maximum or greater to the stated minimum or less.

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TABLE II. Alignment and accuracy requirements.

Characteristic	Permissible tolerance
Base:	
Base straight in longitudinal direction	0.00024"/ft
Base flat or level in transverse direction (no twist permitted)	± 0.00024"/ft
Guideways of workhead and wheelhead parallel with table movement	0.00012"/ft
Workhead:	
Workhead spindle axial slip (in 360 degrees rotation)	0.0004" TIR
Workhead spindle taper runout:	
- 1-1/2 inches from face	0.0005" TIR
- 12 inches from face	0.001" TIR
Workhead spindle periphery and face runout:	
- taper	0.0005" TIR
- face	0.001" TIR
- flange face	0.0005" TIR
Workhead spindle parallel with table travel (in horizontal and vertical planes)	0 to 0.0008"/ft
Workhead spindle parallel with infeed travel of wheel (in vertical plane measured in the 90 degree and 45 degree positions of workhead)	0 to 0.0008"/ft
Workhead and wheelhead spindles on same plane (with wheelhead fully advanced)	0.004"

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TABLE II. Alignment and accuracy requirements - continued,

Characteristic	Permissible tolerance
Wheelhead:	
Wheelhead spindle runout and axial slip (in 360 degrees rotation)	0.0004" TIR
Wheel sleeve flange and periphery runout (in 360 degrees rotation)	0.001" TIR
Internal wheelhead spindle runout and axial slip (in 360 degrees rotation)	0.0004" TIR
Wheelhead spindle parallel with table travel (in horizontal and vertical planes)	0 to 0.0012"/ft
Internal wheel spindle parallel with table travel	0.0012"/ft
Rise or fall of wheelhead spindle in its infeed movement (in total length of infeed motion)	0.002"
Infeed movement of wheelhead slide square with base ways (in total length of infeed motion)	0.0004"
*Table traverse reversal accuracy in 12 inches of table travel at 10 IPM	0.005" TIR
*Rapid infeed stop accuracy: - Coarse - Fine	 0.008" 0.0003"

*When applicable.

Custodians:
activity:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - SM, AT
Navy - MC, YD

Preparing activity:
DLA - IP

Project (3415-0165)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80073C	Grinding Machines, Internal, Universal

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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